

15_Central_Limit_Theorem

Math 742

2026-02-24

Introduction

Suppose you are given 10,000 random samples of size n from some mystery distribution.

For each sample you compute two things:

- sample mean $\bar{x}$
- sample standard deviation s

Now make a confidence interval for the population mean μ using the formula everyone learns in intro stats:

$$\bar{x} \pm 1.96(s/\sqrt{n})$$

(95% confidence)

Now, I simulate this:

```
set.seed(8675309)
B <- 10000 # repetitions
n <- 30
coverage <- numeric(B)

for(i in 1:B) {
  x <- rexp(n, rate = 2) # heavily skewed, mean = 0.5, var = 0.25
  xbar <- mean(x)
  s <- sd(x)
  se <- s / sqrt(n)
  # Check to see if 0.5 (i.e. mean) is in the interval
  coverage[i] <- (0.5 > xbar - 2*se) & (0.5 < xbar + 2*se)
}
mean(coverage)
```

```
## [1] 0.9174
```

Humm... coverage was not 95%? The interval only contained the mean 91.74% of the time.

Maybe $n = 30$ is too small.

Let's try $n = 50, 100, 500$ and 1000.

```
library(dplyr)

set.seed(2025) # reproducible
B <- 20000 # number of replications
n_values <- c(50, 100, 500, 1000)
true_mu <- 0.5
```

```

coverage_results <- numeric(length(n_values))

for (i in seq_along(n_values)) {

  n <- n_values[i]

  # Generate ONE matrix of samples (n rows, B columns)
  samples <- matrix(rexp(n * B, rate = 2), nrow = n)

  # Compute statistics from SAME samples
  xbar <- colMeans(samples)
  s <- apply(samples, 2, sd)

  lower <- xbar - 1.96 * s / sqrt(n)
  upper <- xbar + 1.96 * s / sqrt(n)

  coverage_results[i] <- mean(lower < true_mu & true_mu < upper)
}

tibble(
  n = n_values,
  Coverage = round(100 * coverage_results, 2)
) %>%
  knitr::kable(format = "markdown")

```

n	Coverage
50	92.94
100	93.99
500	95.00
1000	94.80

We already know the Central Limit Theorem says $\sqrt{n}(\bar{X} - \mu)/\sigma \rightarrow N(0, 1)$.

So everyone thinks: “Just plug in s for σ and the interval $\bar{x} \pm 1.96(s/\sqrt{n})$ will work fine for large n .”

But when we had a very skewed distribution this didn’t work out so well.

Even at $n = 1000$ there is still small finite-sample error. Why? Because s is random and variable, and we have no mathematical justification that the ratio $\sigma/s \rightarrow 1$ fast enough, until we learn **Slutsky’s theorem**.

Why we need Slutsky’s Theorem

Slutsky’s theorem says that,

If $X_n \xrightarrow{d} N(0, 1)$ and $Y_n \xrightarrow{p} 1$, then,

$$X_n Y_n \xrightarrow{d} N(0, 1)$$

Think about how this works for the experiment we just investigated.

By the Central Limit Theorem,

$$\frac{\sqrt{n}(\bar{X} - \mu)}{\sigma} \xrightarrow{d} N(0, 1).$$

Also, since $s \xrightarrow{p} \sigma$, we have

$$\frac{\sigma}{s} \xrightarrow{p} 1.$$

Therefore, by Slutsky's theorem,

$$\frac{\sqrt{n}(\bar{X} - \mu)}{s} = \frac{\sqrt{n}(\bar{X} - \mu)}{\sigma} \cdot \frac{\sigma}{s} \xrightarrow{d} N(0, 1).$$

This is why, as n gets larger, the plug-in interval $\bar{X} \pm 1.96 s/\sqrt{n}$ becomes approximately a 95% confidence interval, and the simulated coverage moves closer and closer to 95%.

Mathematical Definitions

Definition 1: Convergence in Probability

A sequence of random variables $X_1, X_2, \dots$ converges in probability to a random variable X if for every $\varepsilon > 0$,

$$\lim_{n \rightarrow \infty} \mathbb{P}(|X_n - X| \geq \varepsilon) = 0$$

or equivalently,

$$\lim_{n \rightarrow \infty} \mathbb{P}(|X_n - X| < \varepsilon) = 1$$

Notice that with this definition we haven't stated that this is an iid sample. The distribution of X_n changes as the subscript changes and the convergence concepts discussed describe different ways in which X_n converges to some limiting distribution as n becomes large.

Definition 2 : Converges almost Surely

A sequence of random variables, $X_1, X_2, \dots$ converges *almost surely* to a random variable X if for every $\varepsilon > 0$,

$$\mathbb{P}(\lim_{n \rightarrow \infty} |X_n - X| < \varepsilon) = 1$$

Notice that this definition and the convergence in probability are almost the same, except for the placement of the limit sign.

Here's an analogy to help you understand this.

Convergence in probability $\rightarrow^P$	Almost sure convergence $\rightarrow^{a.s.}$
Every year on your birthday you throw one new dart from a fresh random position. Each year your throws get closer to the bullseye on average , the probability of missing by more than ε (any $\varepsilon > 0$) goes to zero. Occasional wild throws are allowed , as long as they become rarer and rarer.	Every year you throw one dart and it sticks forever on the board. After infinitely many years, every single dart on the board (except on a set of probability zero) lies arbitrarily close to the bullseye. No more wild throws are allowed — the entire sequence eventually stays inside any ε -ring forever.

Key Takeaway

- Convergence in probability $\xrightarrow{p}$ gets close **most of the time**
- Almost sure convergence $\xrightarrow{as}$ gets close and **stays close forever** (almost surely)

Example: Let $X_1, X_2, \dots$ be iid. The experiment consists of trials. For the n^{th} trial, $\mathbb{P}(X_i = 0) = 1 - 1/n$ and $\mathbb{P}(X_i = 1) = 1/n$

Define our estimator as $\hat{\theta} = X_n$, i.e. the n^{th} observation.

$$\mathbb{E}(\hat{\theta}) = \mathbb{E}(X_n) = 1 \frac{1}{n} + 0 \left(1 - \frac{1}{n}\right) = \frac{1}{n}$$

As $n \rightarrow \infty$ the expected value of the parameter goes to zero.

Thus,

$$\lim_{n \rightarrow \infty} \mathbb{P}(|X_n - 0| > \varepsilon) = \mathbb{P}(X_n = 1) = \frac{1}{n} \rightarrow 0$$

So, $X_n \xrightarrow{p} 0$.

However, X_n does not converge *almost surely*.

Even though this estimator gets close to 0 most of the time, it never truly settles down near 0.

Think of this example as a coin flipping experiment. $\Pr(\text{Head on } n^{\text{th}} \text{ flip}) = 1/n$ and $\Pr(\text{Tail}) = 1 - 1/n$. Although, the probability of getting a head becomes super rare, they will happen. But as n grows they get fewer and fewer this is why this converges in probability to zero.

However, for almost sure convergence getting a head has to stop altogether and this never happens.

Theorem: Weak Law of Large Numbers

Let $X_1, X_2, \dots$ be a iid random variables with $\mathbb{E}(X_i) = \mu$ and $\mathbb{V}(X_i) = \sigma^2 < \infty$. Define $\bar{X}_n = \frac{1}{n} \sum_{i=1}^n X_i$. Then for every $\varepsilon > 0$,

$$\lim_{n \rightarrow \infty} \mathbb{P}(|\bar{X}_n - \mu| < \varepsilon) = 1$$

Proved in an earlier lecture using Chebychev's inequality.

Theorem: Strong Law of Large Numbers

Let $X_1, X_2, \dots$ be a iid random variables with $\mathbb{E}(X_i) = \mu$ and $\mathbb{V}(X_i) = \sigma^2 < \infty$. Define $\bar{X}_n = \frac{1}{n} \sum_{i=1}^n X_i$. Then for every $\varepsilon > 0$,

$$\mathbb{P}\left(\lim_{n \rightarrow \infty} |\bar{X}_n - \mu| < \varepsilon\right) = 1$$

In other words, $\bar{X}_n$ converges *almost surely* to μ .

I won't prove this it's rather involved, uses Kolmogorov's inequality or martingales.

Bottom Line: Everything we do in Math 742 uses the Weak Law (convergence in probability). The weak law is what we use for asymptotics in statistics. The strong law is more important in probability theory and stochastic processes.

Definition 3: Converges in Distribution

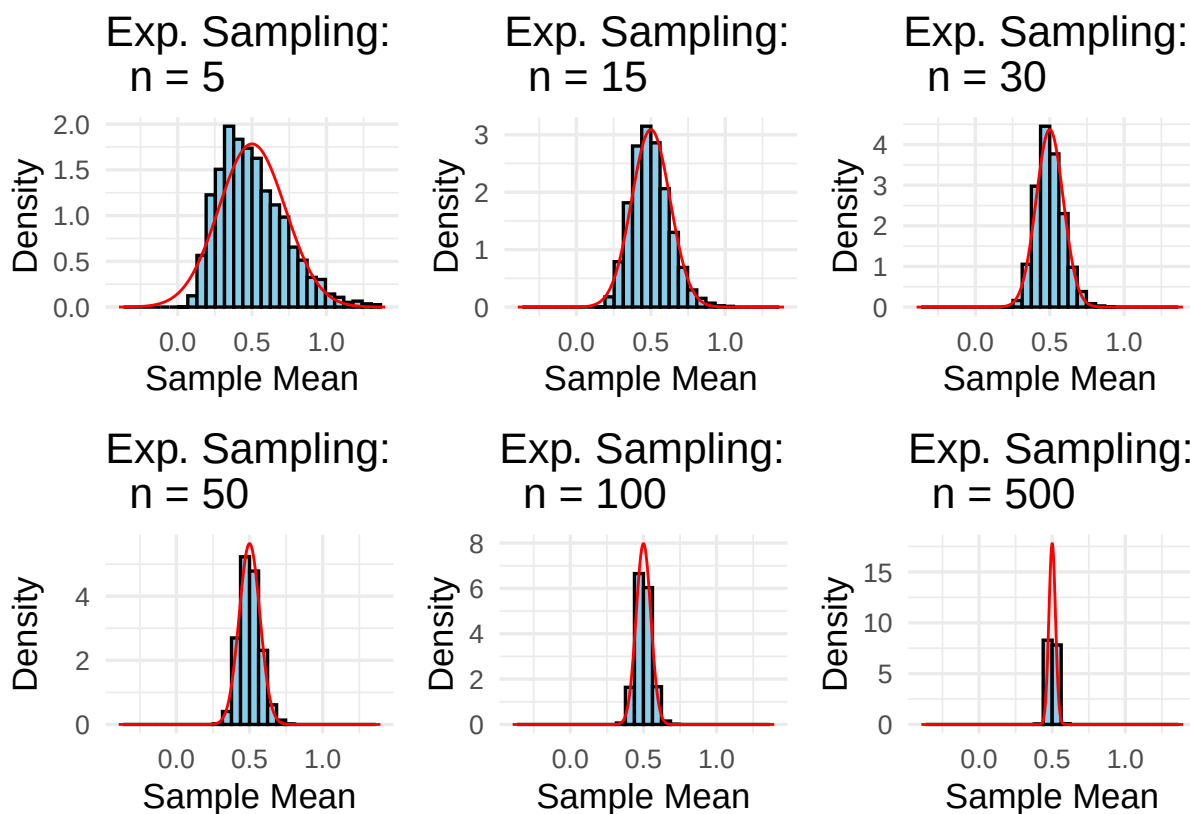
A sequence of random variables $X_1, X_2, \dots$ converges in distribution to a random variable X if

$$\lim_{n \rightarrow \infty} F_{X_n}(x) = F_X(x)$$

at all continuity points of F_X .

What this means: When n gets huge, if you take a million samples of X_n and draw a histogram, it becomes impossible to tell apart from the histogram you'd get from the limiting random variable X . The whole probability “picture” settles down even though each individual X_n might be completely different from X .

This is convergence in distribution in action!



Theorem: If a sequence of random variables $X_1, X_2, \dots$ converges in probability to a constant μ iff the sequence converges in distribution to μ .

Why this matters

- Sample mean: $\bar{X}_n \xrightarrow{P} \mu$ (by WLLN)
- Therefore $\bar{X}_n \xrightarrow{d} \mu$
- Then apply CLT + Slutsky $\rightarrow \sqrt{n}(\bar{X}_n - \mu)/s_n \xrightarrow{d} N(0, 1)$
- $\rightarrow$ all large-sample inference works!

Central Limit Theorem:

Let $X_1, X_2, \dots$ be i.i.d. with $\mathbb{E}[X_i] = \mu$, $\text{Var}(X_i) = \sigma^2 < \infty$.

Then,

$$\frac{\sqrt{n}(\bar{X}_n - \mu)}{\sigma} \xrightarrow{d} N(0, 1) \quad \text{as } n \rightarrow \infty.$$

Proof using moment generating functions

1. Center and scale the variables:

$$Z_i = \frac{X_i - \mu}{\sigma} \implies \mathbb{E}[Z_i] = 0, \text{Var}(Z_i) = 1.$$

Then,

$$\sqrt{n}(\bar{X}_n - \mu) = \frac{1}{\sqrt{n}} \sum_{i=1}^n (X_i - \mu) = \frac{1}{\sqrt{n}} \sum_{i=1}^n \sigma Z_i = \sigma \cdot \frac{1}{\sqrt{n}} \sum_{i=1}^n Z_i$$

Let $\sigma \cdot \frac{1}{\sqrt{n}} \sum_{i=1}^n Z_i = \sigma W_n$.

It suffices to show $W_n \xrightarrow{d} N(0, 1)$.

2. Assume the MGF exists in a neighborhood of 0.

By a second-order Taylor expansion using $\mathbb{E}[Z_i] = 0$ and $\mathbb{V}(Z_i) = 1$:

$$M_{Z_i}(t) = \mathbb{E}[e^{tZ_i}] = 1 + t \cdot 0 + \frac{t^2}{2} \cdot 1 + o(t^2) = 1 + \frac{t^2}{2} + o(t^2) \quad \text{as } t \rightarrow 0.$$

3. The MGF of the sum $S_n = \sum_{i=1}^n Z_i$ is

$$M_{S_n}(t) = \mathbb{E}\left[e^{t \sum_{i=1}^n Z_i}\right] = \mathbb{E}\left[\prod_{i=1}^n e^{tZ_i}\right].$$

Since the Z_i 's are independent,

$$\mathbb{E}\left[\prod_{i=1}^n e^{tZ_i}\right] = \prod_{i=1}^n \mathbb{E}[e^{tZ_i}].$$

Because the Z_i 's are identically distributed,

$$\mathbb{E}[e^{tZ_i}] = M_Z(t) \quad \text{for every } i.$$

Thus,

$$M_{S_n}(t) = [M_Z(t)]^n.$$

So the MGF of $W_n = S_n/\sqrt{n}$ is,

$$M_{W_n}(t) = \mathbb{E}\left[e^{tS_n/\sqrt{n}}\right] = (M_Z(t/\sqrt{n}))^n.$$

$$M_{W_n}(t) = \mathbb{E}\left[e^{tS_n/\sqrt{n}}\right] = M_{S_n}(t/\sqrt{n}) = (M_Z(t/\sqrt{n}))^n.$$

4. Plug into the expansion (i.e. plug in $t/\sqrt{n}$ for t):

$$M_Z(t/\sqrt{n}) = 1 + \frac{(t/\sqrt{n})^2}{2} + o(1/n) = 1 + \frac{t^2}{2n} + o(1/n).$$

Note: $o\left(\left(\frac{t}{\sqrt{n}}\right)^2\right) = o(1/n)$ Since t is fixed, and $n \rightarrow \infty \left(\frac{t}{\sqrt{n}}\right)^2 \sim 1/n$.

5. Let $n \rightarrow \infty$.

$$M_{W_n}(t) = \left(1 + \frac{t^2}{2n} + o(1/n)\right)^n.$$

Let

$$a_n = \frac{t^2}{2n} + r_n, \quad \text{where } r_n = o(1/n).$$

Define

$$x_n = na_n = \frac{t^2}{2} + nr_n.$$

Since $r_n = o(1/n)$, we have $nr_n \rightarrow 0$, and therefore

$$x_n \rightarrow \frac{t^2}{2}.$$

Now observe

$$(1 + a_n)^n = \left(1 + \frac{x_n}{n}\right)^n.$$

Using the fact that if $x_n \rightarrow x$, then

$$\left(1 + \frac{x_n}{n}\right)^n \rightarrow e^x,$$

we conclude

$$M_{W_n}(t) \rightarrow e^{t^2/2}.$$

6. The limit $e^{t^2/2}$ is exactly the MGF of a standard normal $N(0, 1)$.

By the continuity theorem for MGFs,

$$W_n \xrightarrow{d} N(0, 1) \implies \sqrt{n}(\bar{X}_n - \mu) \xrightarrow{d} N(0, \sigma^2). \square$$

Note: The continuity theorem for MGFs tells us that if the MGFs converge to the MGF of a known distribution (in a neighborhood of 0), then the random variables converge in distribution to that distribution.

But the CLT is not magic, it has limits.

Here's the **sample median** from a Cauchy distribution (no mean, infinite variance) shown earlier.

n	Ratio MC Variance/Asymptotic Variance
5	2.44
25	1.14
100	1.02
500	1.006

Even at $n = 500$, the normal approximation is still slightly optimistic, because the CLT requires finite variance. Break that assumption and convergence crawls.

Today we've proved why the CLT works when the conditions hold.

Bottom line for Math 742

Everything we do after this (Wald intervals, delta method, asymptotic normality of MLEs, t-statistics with s instead of σ , etc.) rests on this single theorem.

Next, we look at Slutsky's theorem and what it does to save our confidence intervals when we plug in s instead of σ .

Slutsky's Theorem (the version we actually use every day)

Let $X_n \xrightarrow{d} X$ and $Y_n \xrightarrow{p} c$ (where c is a constant).

Then

1. $X_n + Y_n \xrightarrow{d} X + c$
2. $X_n \cdot Y_n \xrightarrow{d} X \cdot c$

(in particular, if $c \neq 0$, then $X_n/Y_n \xrightarrow{d} X/c$)

Why this saves our confidence intervals

In real life we compute

$$T_n = \sqrt{n} \frac{\bar{X}_n - \mu}{s_n} \quad \text{instead of} \quad \sqrt{n} \frac{\bar{X}_n - \mu}{\sigma}$$

We can write

$$T_n = \underbrace{\sqrt{n} \frac{\bar{X}_n - \mu}{\sigma}}_{\xrightarrow{d} N(0,1)} \cdot \underbrace{\frac{\sigma}{s_n}}_{\xrightarrow{p} 1}$$

So Slutsky says: $T_n \xrightarrow{d} N(0,1) \cdot 1 = N(0,1)$

→ the plug-in interval $\bar{X}_n \pm 1.96 \frac{s_n}{\sqrt{n}}$ is asymptotically valid!

Proof

We only prove the multiplication case (the one we need).

Let $X_n \xrightarrow{d} X$ and $Y_n \xrightarrow{p} c$.

→ Because convergence in distribution is **defined through expectations of test functions**.

In order to prove convergence in distribution, we use the Portmanteau characterization:

$X_n \xrightarrow{d} X$ if and only if $\mathbb{E}[f(X_n)] \rightarrow \mathbb{E}[f(X)]$ for every bounded continuous function f .

Now, fix any bounded continuous function $f : \mathbb{R} \rightarrow \mathbb{R}$.

We want to show $\mathbb{E}[f(X_n Y_n)] \rightarrow \mathbb{E}[f(Xc)]$.

Since $Y_n \rightarrow c$ in probability, for any $\varepsilon > 0$,

$$\mathbb{P}(|Y_n - c| > \varepsilon) \rightarrow 0 \text{ as } n \rightarrow \infty.$$

Split the expectation into two parts:

$$\mathbb{E}[f(X_n Y_n)] = \mathbb{E}[f(X_n Y_n) \cdot \mathbf{1}_{|Y_n - c| \leq \varepsilon}] + \mathbb{E}[f(X_n Y_n) \cdot \mathbf{1}_{|Y_n - c| > \varepsilon}]$$

The multiplier Y_n is arbitrarily close to the constant c with probability approaching 1.

On that high-probability event, the product $X_n Y_n$ differs only slightly from $X_n c$. Because f is bounded and continuous, small changes in its argument produce small changes in its value, so $f(X_n Y_n)$ is close to $f(X_n c)$ with high probability. The contribution from the complementary low-probability event vanishes because f is bounded.

Meanwhile, since $X_n \xrightarrow{d} X$ and $f(\cdot c)$ is bounded and continuous,

$$\mathbb{E}[f(X_n c)] \rightarrow \mathbb{E}[f(Xc)].$$

Combining these facts,

$$\mathbb{E}[f(X_n Y_n)] \rightarrow \mathbb{E}[f(Xc)].$$

Since this holds for every bounded continuous f , we are done by the portmanteau theorem.

Convergence in probability means Y_n eventually lives in any tiny ball around c with probability $\rightarrow 1$.

Convergence in distribution only cares about the shape of X_n . Multiplying by something that is almost surely a constant doesn't mess up the shape, it just slides or stretches it by a fixed amount.

Python Code

```
import numpy as np

np.random.seed(8675309)
B = 10000
n = 30
coverage = np.zeros(B, dtype=bool)

for i in range(B):
    x = np.random.exponential(scale=1/2, size=n) # rate=2 + scale=1/2
    xbar = np.mean(x)
    s = np.std(x, ddof=1) # R uses sample sd by default (ddof=1)
    se = s / np.sqrt(n)

    coverage[i] = (0.5 > xbar - 2*se) and (0.5 < xbar + 2*se)

np.mean(coverage)
```

```

## np.float64(0.9203)
import numpy as np
import pandas as pd

np.random.seed(2025)

B = 20000
n_values = [50, 100, 500, 1000]
true_mu = 0.5

coverage_results = []

for n in n_values:
    # Generate all samples in one matrix: shape (n, B)
    samples = np.random.exponential(scale=1/2, size=(n, B)) # rate=2 + scale = 1/2

    xbar = samples.mean(axis=0)
    s = samples.std(axis=0, ddof=1)

    lower = xbar - 1.96 * s / np.sqrt(n)
    upper = xbar + 1.96 * s / np.sqrt(n)

    coverage = np.mean((lower < true_mu) & (true_mu < upper))
    coverage_results.append(coverage)

# Table
df = pd.DataFrame({
    "n": n_values,
    "Coverage (%)": np.round(100 * np.array(coverage_results), 2)
})

df

```

```

import numpy as np
import matplotlib.pyplot as plt
from scipy.stats import norm

# sample sizes
n_values = [5, 15, 30, 50, 100, 500]

# population distribution: Exponential with rate lambda = 2
lambda_ = 2
mu = 1 / lambda_ # mean = 0.5
sigma = 1 / lambda_ # sd = 0.5

# number of simulations
B = 5000

# determine common x-range (based on n = 5)
sd5 = sigma / np.sqrt(5)
xmin = mu - 4 * sd5
xmax = mu + 4 * sd5
xgrid = np.linspace(xmin, xmax, 500)

```

```

# 2x3 subplot grid
fig, axes = plt.subplots(2, 3, figsize=(14, 8))
axes = axes.flatten()

np.random.seed(123)

for i, n in enumerate(n_values):
    ax = axes[i]

    # simulate B sample means
    samples = np.random.exponential(scale=1/lambda_, size=(B, n))
    sample_means = samples.mean(axis=1)

    # histogram of sample means
    ax.hist(sample_means, bins=30, density=True, edgecolor='black')

    # theoretical normal
    ax.plot(xgrid, norm.pdf(xgrid, loc=mu, scale=sigma/np.sqrt(n)), linewidth=1.2)

    ax.set_xlim(xmin, xmax)
    ax.set_title(f"Exp. Sampling:\n n = {n}")
    ax.set_xlabel("Sample Mean")
    ax.set_ylabel("Density")

plt.tight_layout()
plt.show()

```

